

Non-Disclosure and Rental Market Tension

Descriptive Evidence

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Sciences Po



SciencesPo.

The data: four ingredients

(i) Demand: seeker activity

Leads / inquiries a listing receives from housing seekers; we use the first-week count. Unobserved when no lead data exists, and is treated as missing, not zero.

(iii) Listing characteristics

Room count (T1–T4+), construction year, advertised DPE class and the coarse EPC band, asking rent per m².

(ii) Liquidity: time on market

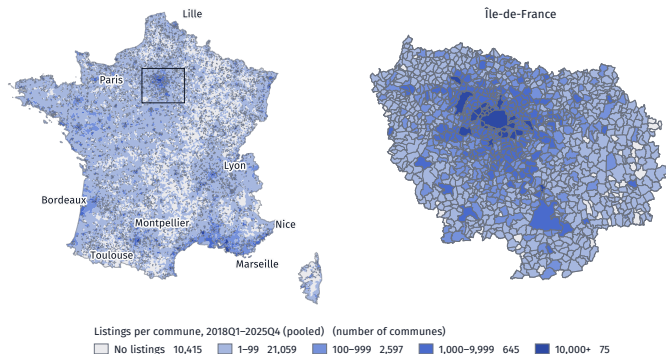
Days from a listing going live to leaving the platform: how fast the local market clears. Trimmed at the pooled 99th percentile.

(iv) Tension scheme: the law

Zone tendue defined by décret no. 2013-392 of 10 May 2013; rent-level control added by loi ELAN no. 2018-1021, art. 140 (République française 2013, 2018). Four levels: rent-level cap, rent-increase cap, vacancy-tax only, not tight.

RENTAL-TENSION listing-spell panel, France, 2018Q1–2025Q4 (≈ 6.7 M spells), aggregated to the listing-start quarter; outcomes trimmed to the pooled 1st–99th percentile. Dashed line = July 2021 DPE reform (2021Q3).

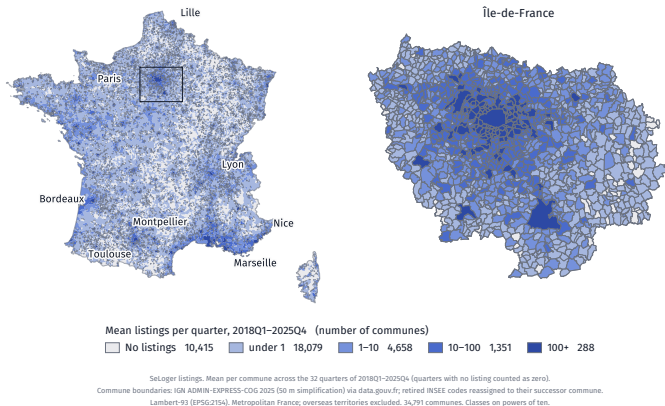
Pooled over eight years, coverage looks near-universal across France



SeLoger listings, 2018Q1–2025Q4, pooled.
Commune boundaries: IGN ADMIN-EXPRESS-COG 2025 via data.gouv.fr; retired INSEE codes reassigned to their successor commune.
Lambert-93 (EPSG:2154). Metropolitan France; overseas territories excluded. 34,791 communes. Classes on powers of ten.

Pooled listing count, 2018Q1–2025Q4: 70% of communes have at least one listing.

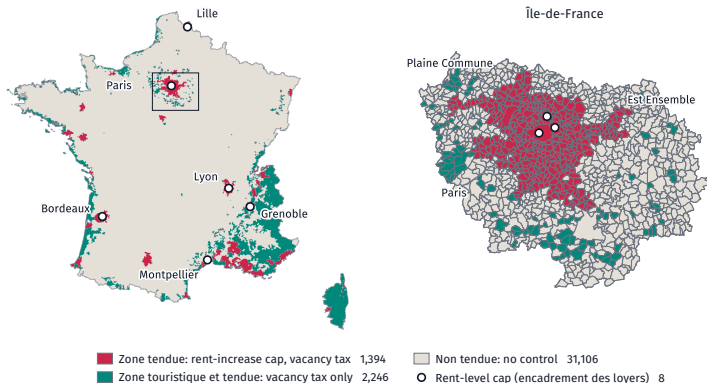
Quarter to quarter, the flow is thin: half of covered communes see under one listing



Mean listings per quarter conflates market size with coverage: a commune with few listings may be small, not under-covered.

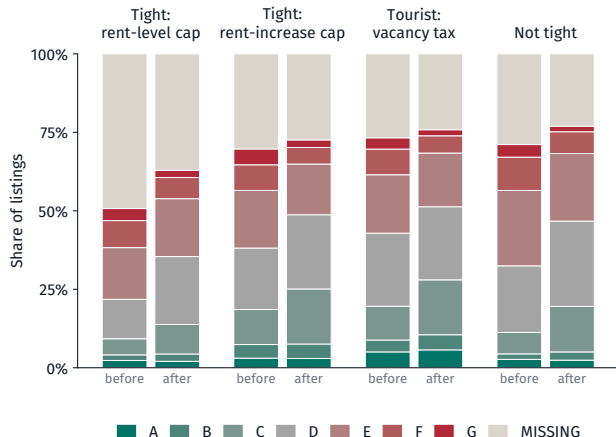
What Tension Zones Look Like

Tension zones concentrate in Île-de-France, the big agglomerations and the tourist coasts



TLV zoning 2023: décret n° 2013-392 of 10 May 2013 (art. 232 CGI), reformed by décret n° 2023-822 of 25 August 2023.
Source: data.gouv.fr / Ministère de la Transition écologique. Commune boundaries: IGN ADMIN-EXPRESS-COG 2025 (50 m simplification, data.gouv.fr).
Rent-level control: loi ELAN n° 2018-1021, art. 140. Lambert-93 (EPSG:2154). Metropolitan France; overseas territories excluded. 34,746 communes.

In the tightest zone, over a third of listings still publish with no EPC despite mandatory disclosure



No-EPC share

	Before	After	Δ
Level cap	49%	37%	−12 pp
Increase cap	30%	27%	−3 pp
Tourist	27%	24%	−3 pp
Not tight	29%	23%	−6 pp

Listings with no advertised EPC, before / after

1 July 2021.

Tight zones concentrate many listings in very few communes

Tension zone	Before 2021-07		After 2021-07	
	Listings	Communes	Listings	Communes
Tight: rent-level cap	607 612	68	762 621	69
Tight: rent-increase cap	1 213 272	1 322	1 624 742	1 322
Tourist: vacancy tax	64 113	1 523	73 990	1 589
Not tight	853 431	17 460	1 025 770	18 279

Before = 2018-01 to 2021-06 (42 months); after = 2021-07 to 2025-12 (54 months); counts are not directly comparable across periods. Communes are distinct INSEE codes; room-count detail is in the figure above.

Tension is an administrative designation, not an observable characteristic

The question

The four zones are drawn by decree, not by a continuous variable. Do they proxy for deeper compositional differences, or does the legal boundary carve through otherwise similar communes?

Approach

Merge the listing panel's tension classification onto the 2025 commune frame (34 746 metropolitan communes), then compare population-weighted means of five composition variables across the four zones.

Five composition variables

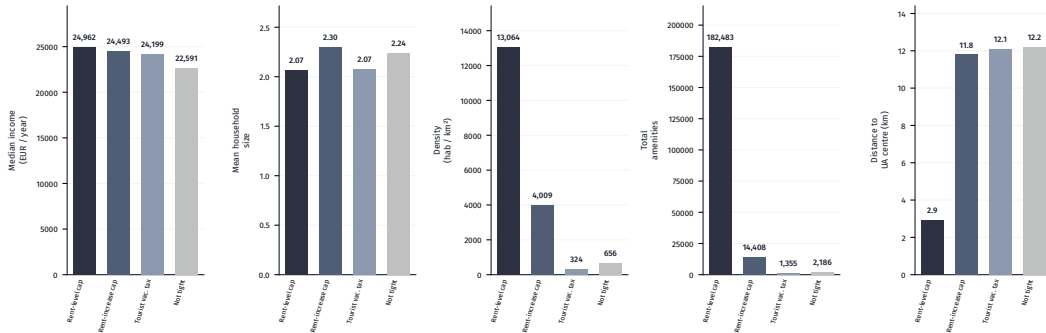
- Median standard of living (FiLoSoFi 2021)
- Mean persons per fiscal household
- Population density (hab / km²)
- Total amenities (BPE 2025)
- Distance to urban-area centre (AAV 2020)

Two weightings

Population-weighted (main) and unweighted (appendix), because the rent-level-cap zone has only 68 communes but 12% of the population.

Tension zones differ sharply on density and amenities, less so on income

Commune composition by tension zone (population-weighted means)



Population-weighted commune means. Source: FiLoSoFi 2021, BPE 2025, AAV 2020, IGN.

Composition summary: population-weighted zone means

	Level cap	Increase cap	Tourist	Not tight
Median income (€/yr)	24 962	24 493	24 199	22 591
Household size	2.07	2.30	2.07	2.24
Density (hab/km ²)	13 064	4 009	324	656
Total amenities	182 483	14 408	1 355	2 186
Distance to UA centre (km)	2.9	11.8	12.1	12.2
Communes	68	1 324	2 244	31 110

Population-weighted means over 34 746 metropolitan communes. Source: FiLoSoFi 2021, BPE 2025, AAV 2020, IGN.

The zones partition along the urban gradient, not along income

What separates the zones

- Density: level-cap communes are 20× denser than not-tight
- Amenities: 130× more equipment in level-cap zones
- Distance to UA centre: 3 km (level cap) vs. 12 km (not tight)
- These communes are city centres

What does not separate them

- Income spread is only €2 400/yr across all four zones
- About one-tenth of the national interquartile range
- Household size similarly compressed (2.07–2.30)

Implication for identification

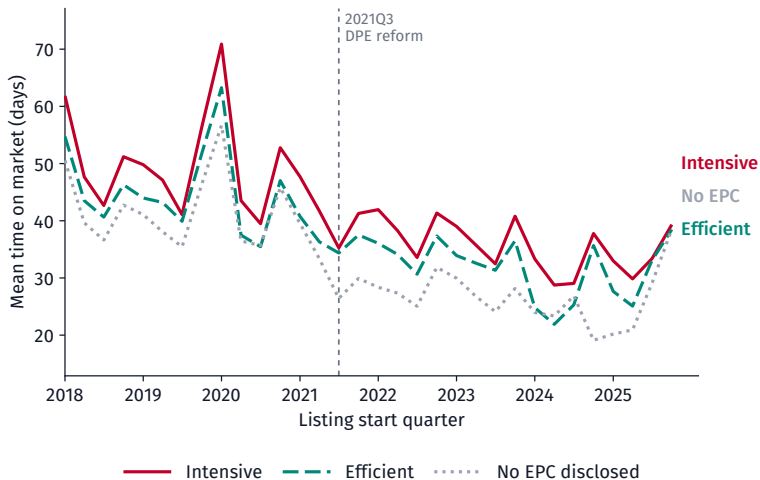
- Cross-zone comparisons $\neq$ “rich vs. poor”
- They compare urban-core markets (many outside options, thick services) against suburban and rural ones

Unweighted check

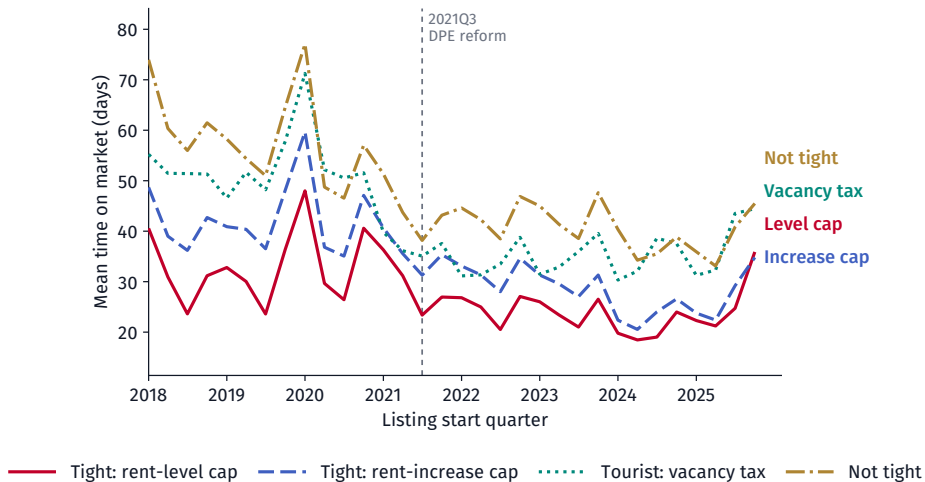
- Dropping population weights reverses the income ranking
- Rent-increase-cap communes richest (€26 700); level-cap poorer (€23 615)
- Headline pattern driven by Paris and Lyon, not the typical commune

Time on Market

No-EPC listings clear systematically faster, despite non-compliance

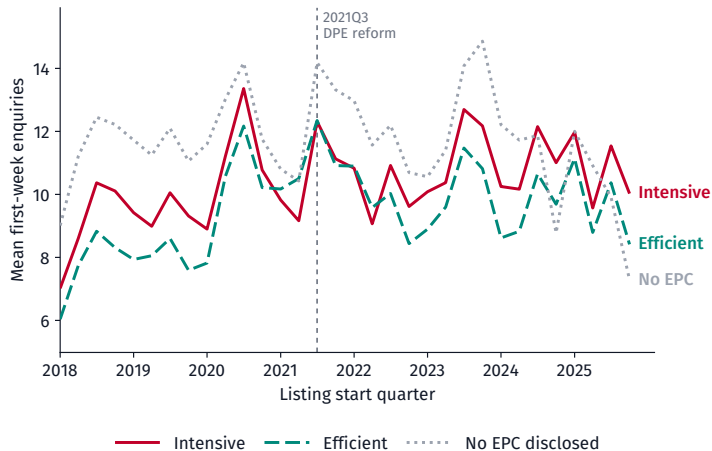


Tighter zones let faster



First-Week Enquiries

Even under mandatory disclosure, non-compliant listings draw about 2 more seeker leads than efficient ones

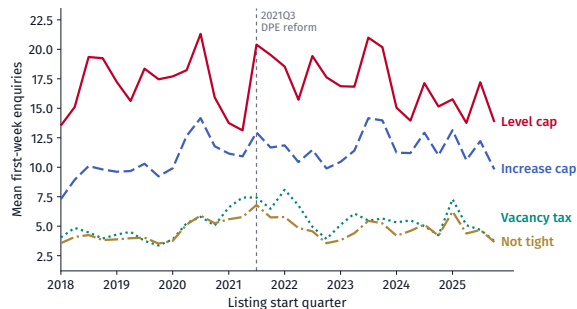


Mean first-week enquiries

	Before Jul '21	After Jul '21
Intensive	9.6	10.8
Efficient	8.9	9.9
No EPC	11.5	11.9

Observed leads only, trimmed at the pooled 99th percentile. Split at 1 July 2021.

Seeker demand is strongest in the rent-level-cap zone, but grows the least after the reform

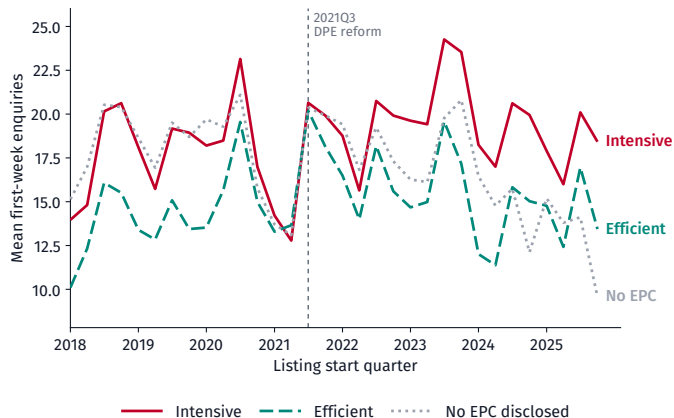


Mean first-week enquiries

	Before	After	Δ
Level cap	16.5	17.0	+3%
Increase cap	10.3	11.7	+13%
Tourist	4.8	5.5	+15%
Not tight	4.5	4.9	+8%

Observed leads only, trimmed at the pooled
99th percentile. Split at 1 July 2021.

In the rent-level-cap zone, demand growth favours intensive listings; no-EPC demand is flat

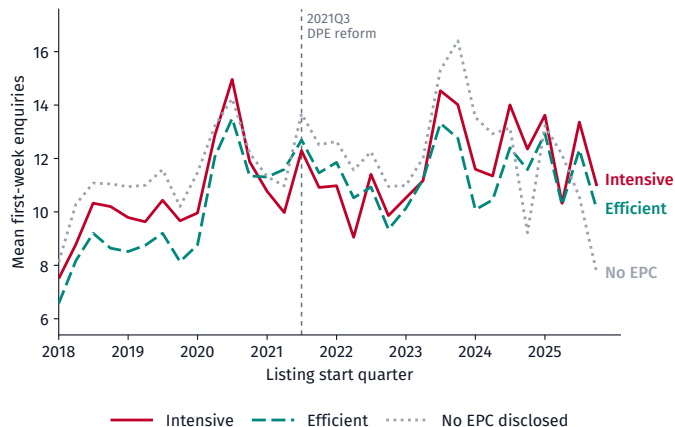


Mean first-week enquiries

	Before	After	Δ
Intensive	17.1	19.3	+13%
Efficient	14.1	15.2	+7%
No EPC	17.2	17.1	-0%

Tight: rent-level cap. Observed leads only, trimmed at the pooled 99th percentile.

In the rent-increase-cap zone, demand grows across every EPC band, most for efficient listings

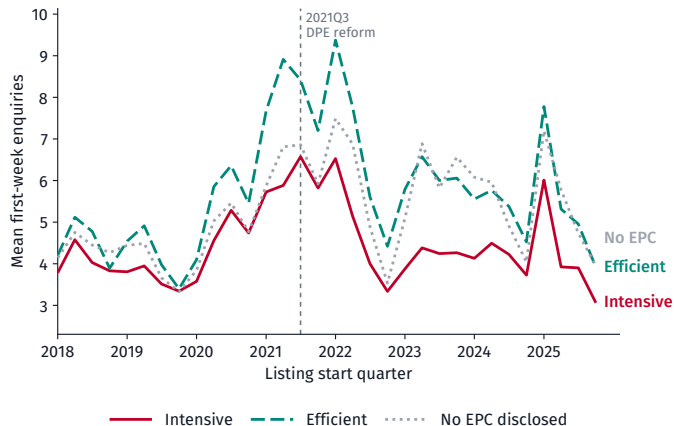


Mean first-week enquiries

	Before	After	Δ
Intensive	10.3	11.7	+14%
Efficient	9.7	11.4	+18%
No EPC	11.2	12.3	+10%

Tight: rent-increase cap. Observed leads only, trimmed at the pooled 99th percentile.

In tourist zones, demand for undisclosed listings grows fastest, closing in on efficient ones

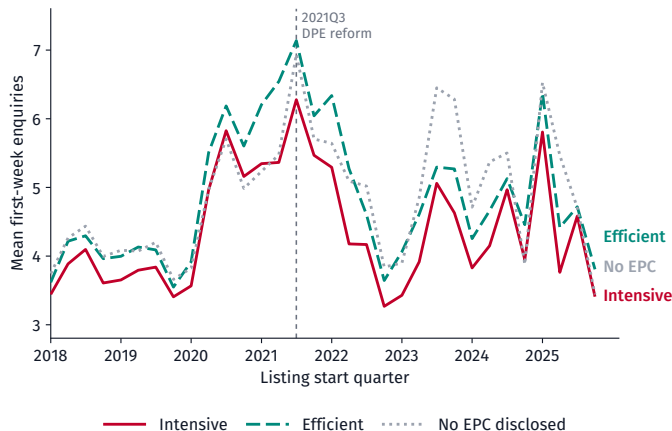


Mean first-week enquiries

	Before	After	Δ
Intensive	4.3	4.5	+5%
Efficient	5.2	5.9	+13%
No EPC	4.8	5.8	+21%

Tourist: vacancy tax. Observed leads only, trimmed at the pooled 99th percentile.

In non-tight zones, no-EPC listings see the largest demand gain and pull ahead after 2021



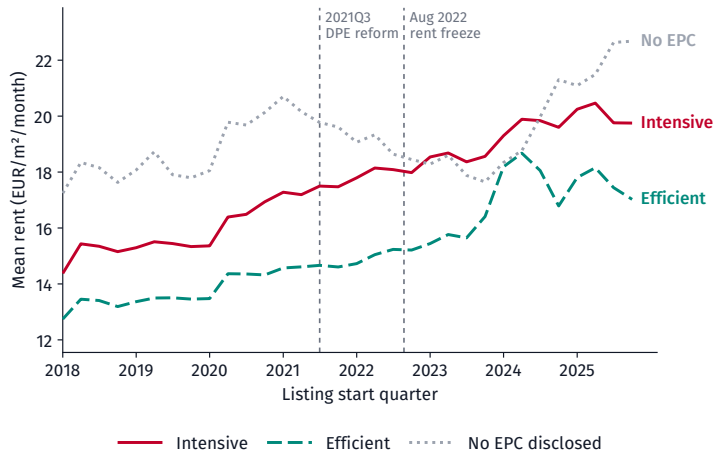
Mean first-week enquiries

	Before	After	Δ
Intensive	4.2	4.4	+5%
Efficient	4.7	4.9	+4%
No EPC	4.6	5.3	+16%

Not a tight zone. Observed leads only, trimmed at the pooled 99th percentile.

Asking Rent per m²

Enforcement of the rent freeze is failing: intensive-property rents keep rising after August 2022

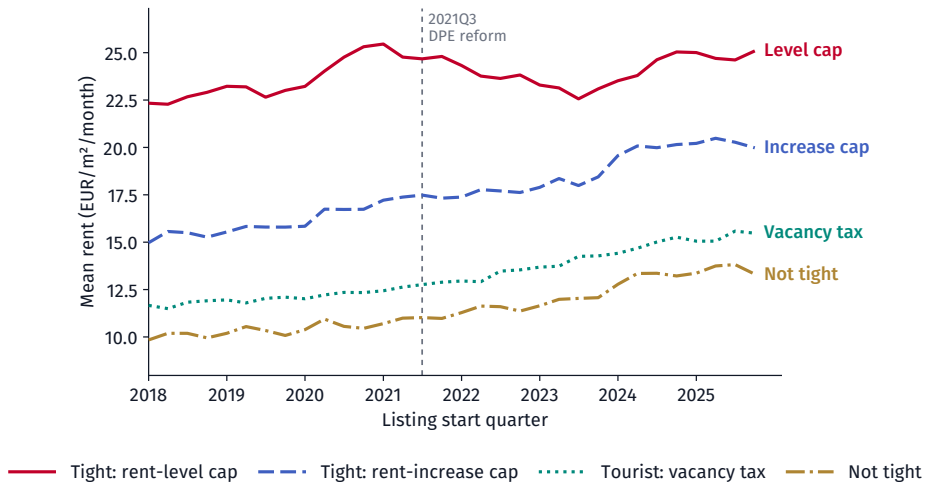


Rent per m² since the freeze

	Aug '22 €/m ² to 2025Q4	
Intensive	18.1	+9.2%
Efficient	15.2	+11.7%
No EPC	18.6	+21.7%

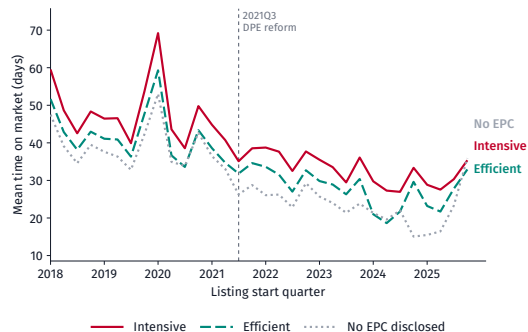
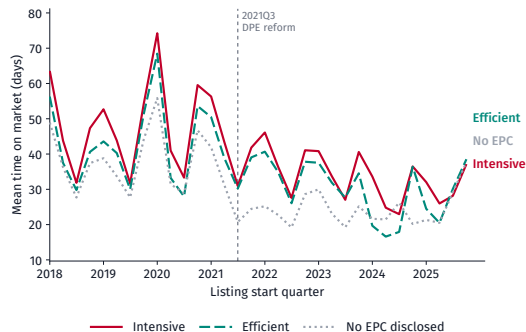
Quarterly mean, trimmed at the pooled 99th percentile. Rent freeze enforced 24 Aug 2022; intensive-property rents should have stagnated.

Rent-controlled zones carry the highest €/m²; all zones trend up



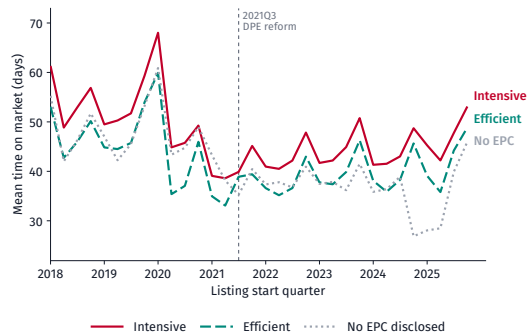
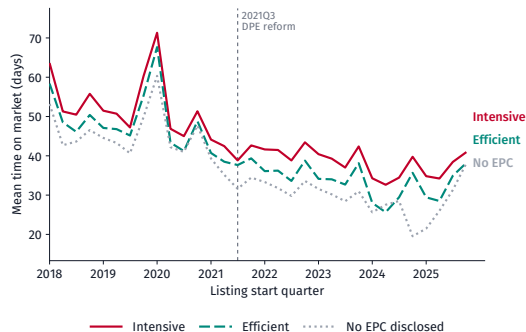
By Dwelling Size

Time on market: the EPC ordering holds across dwelling sizes (T1, T2)



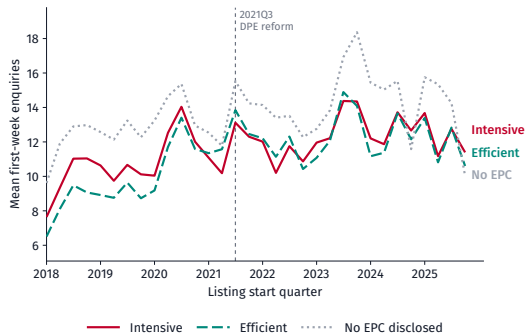
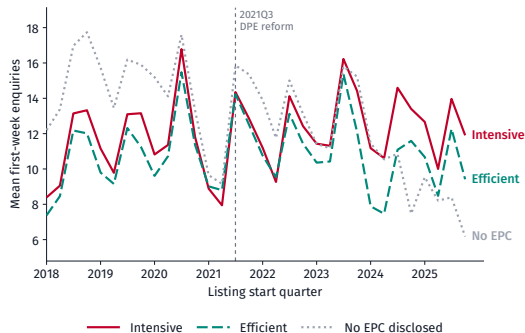
Quarterly mean days on market, by EPC band. Left: 1 room. Right: 2 rooms. Own scale each.

Time on market: the EPC ordering holds across dwelling sizes (T3, T4+)



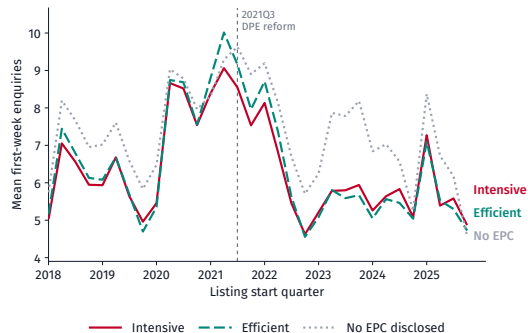
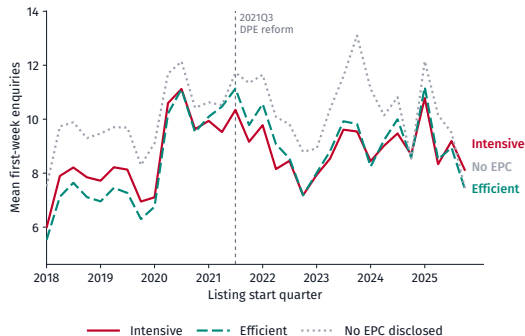
Quarterly mean days on market, by EPC band. Left: 3 rooms. Right: 4 or more. Own scale each.

First-week enquiries by EPC band across dwelling sizes (T1, T2)



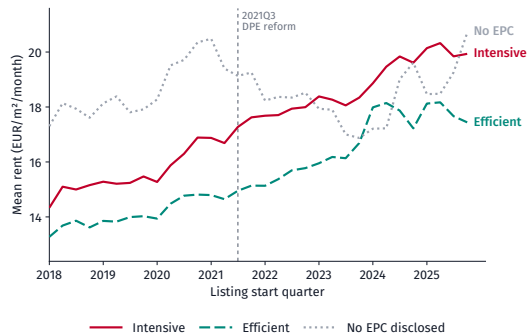
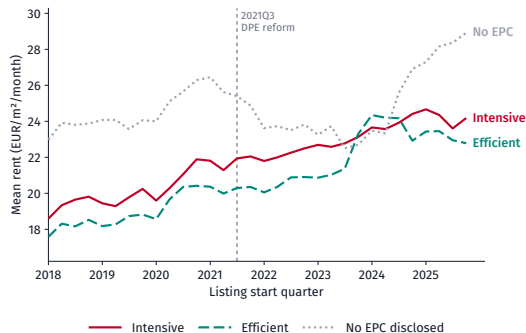
Quarterly mean first-week enquiries. Left: 1 room. Right: 2 rooms. Own scale each.

First-week enquiries by EPC band across dwelling sizes (T3, T4+)



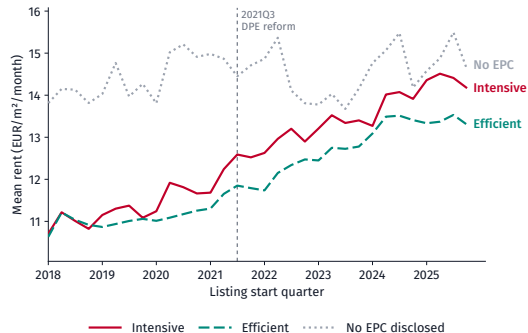
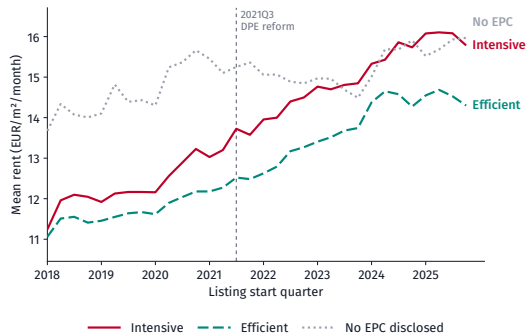
Quarterly mean first-week enquiries. Left: 3 rooms. Right: 4 or more. Own scale each.

Smaller units carry a higher €/m², but the EPC ordering is the same (T1, T2)



Quarterly mean €/m²/month, by EPC band. Left: 1 room. Right: 2 rooms. Own scale each.

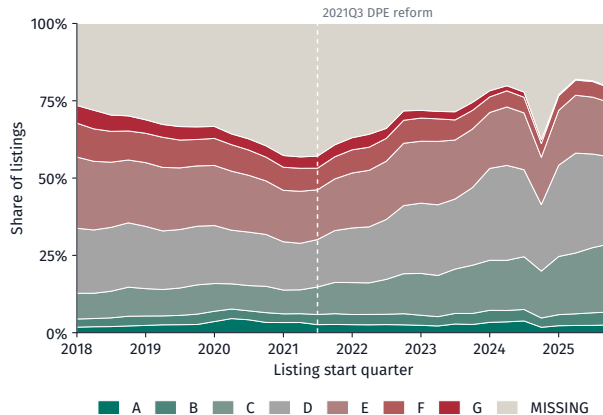
Smaller units carry a higher €/m², but the EPC ordering is the same (T3, T4+)



Quarterly mean €/m²/month, by EPC band. Left: 3 rooms. Right: 4 or more. Own scale each.

DPE Disclosure Over Time

Publishing without an EPC persists: still about one listing in five in 2025



Quarterly DPE composition of all listings; the grey MISSING band is no advertised EPC.

Composition-Adjusted Indices

Everything so far is a raw average; from here on, one common property held fixed

Twelve cells, twelve regressions

4 tension zones $\times$ 3 EPC bands. One regression per cell, *no pooling*: every coefficient in this section carries an implicit (z, b) subscript.

One common reference property

Two rooms, floors 1–3, construction year not advertised, unfurnished, no lift, non-electric heating, 45 m², in the zone's modal commune. Every reference level is *dropped* from the design and log area is centred at log 45, so the intercept *is* that property's prediction.

Standard errors

insee_code-clustered throughout, combined across the three rotations; shaded bands are ± 1 SE.

Quarterly time axis

Three phase-shifted quarter grids (starting Jan / Feb / Mar 2018), each fit as a separate regression and averaged. Every calendar month is directly estimated, so nothing is recovered by interpolation.

Warning: Tension zones *partition* communes, so each zone necessarily uses its own reference commune: a cross-zone comparison of *levels* mixes the regime with that location. Read the within-zone band gap instead.

What the Three Models Say

Key takeaways from the reduced-form estimates

Market	Price hedonic log-OLS	Demand NB2 count	Clearing pace cloglog hazard
Rent-level cap	• Transitory shock, reverts	• No gap between labels	• No difference
Rent-increase cap	▼ Undisclosed priced as intensive	▼ Strong penalty on intensive	• No difference
Tourist, vacancy tax	▲ Non-disclosure premium	• No gap between labels	• No difference
No rent control	• EPC not priced in	▼ Slight penalty on intensive	▼ Intensive clears slower

▼ significant penalty ▲ significant premium for non-disclosure • no significant difference

Model 1: Asking Rent, Hedonic log-OLS

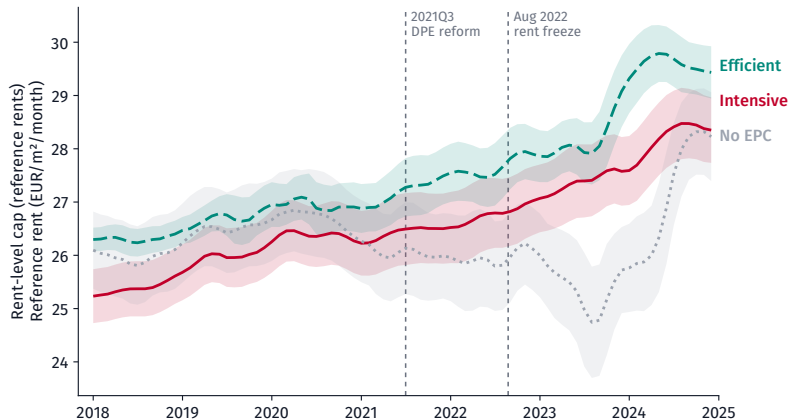
Model 1: a hedonic rent index, estimated separately in each zone × band cell

$$\log p_i = \alpha + \beta_{\text{area}} \log\left(\frac{\text{area}_i}{45}\right) + X_i' \beta + \sum_q \delta_q \mathbf{1}[q(t_i) = q] + \theta_{c(i)} + \varepsilon_i$$

Notation

- p_i : asking rent per m² per month, listing i
- X_i : rooms, floor band, build-year band, furnished, lift, electric heating. Every “missing” is its own modelled category, never imputed
- δ_q : quarter-block effect, **the index**
- $\theta_{c(i)}$: commune dummy for every `insee_code` with > 30 spells in the cell; the rest pool into `commune_other`

The shock appears only transitory: the no-EPC index diverges from the reform's intended trajectory, then reverts

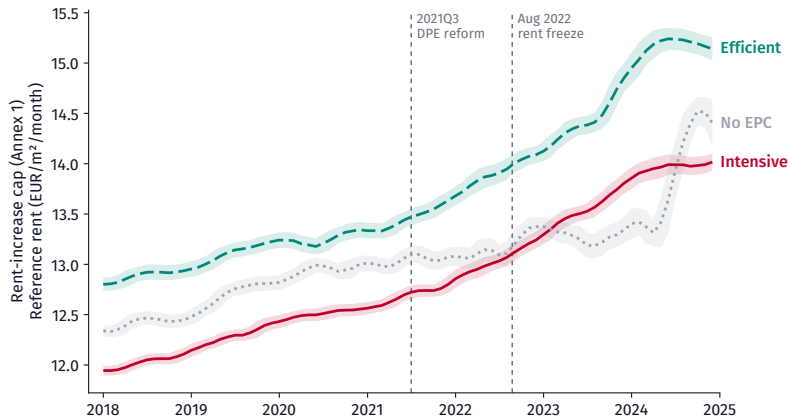


No-EPC level and gaps to it, €/m²/month

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	26.3	26.1	26.2
Intensive	-0.4	+1.2	+1.4
Efficient	+0.4	+2.2	+2.5

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

Textbook Grossman and Milgrom: undisclosed dwellings are penalised and priced at the energy-intensive level

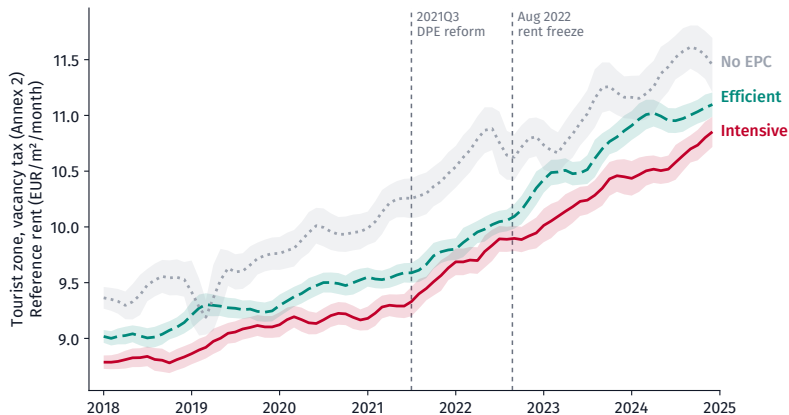


No-EPC level and gaps to it, €/m²/month

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	12.7	13.4	13.5
Intensive	-0.4	0.0	+0.1
Efficient	+0.4	+1.0	+1.1

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

The non-disclosure premium here aligns with Patras (2024, 2026)

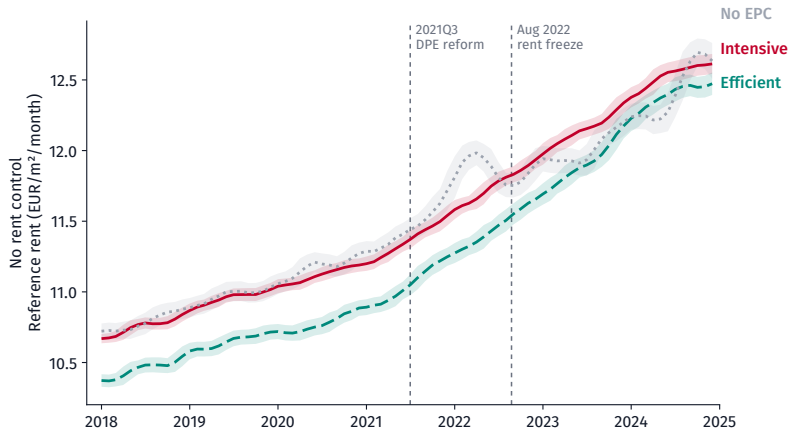


No-EPC level and gaps to it, €/m²/month

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	9.7	10.9	11.1
Intensive	-0.7	-0.8	-0.8
Efficient	-0.4	-0.5	-0.4

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

No significant price gradient: this market does not price the EPC, plausibly too few EPC-sensitive renters

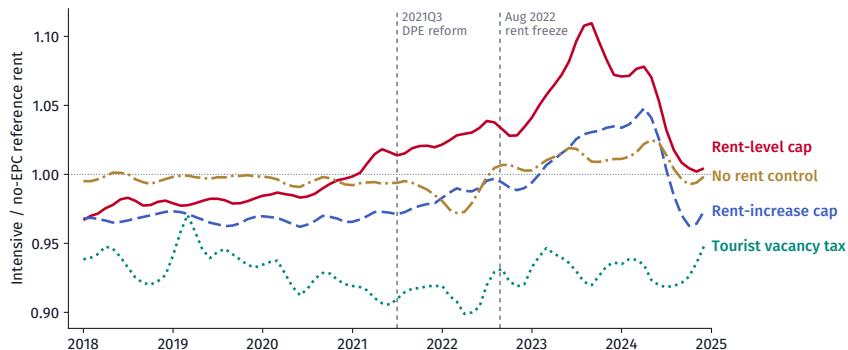


No-EPC level and gaps to it, €/m²/month

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	11.0	12.0	12.2
Intensive	0.0	0.0	+0.1
Efficient	-0.4	-0.2	-0.1

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

Strong heterogeneous effect of EPC disclosure as a function of market tension



Ratio of the reference property's rent, intensive over no-EPC, inside each zone. At 1 the market prices a withheld certificate exactly as if it were E-G. Both bands share the zone's reference commune, so this ratio is comparable across zones.

Takeaway: the price response to disclosure changes sign across the four regimes

Rent-level cap

The reform shock is transitory. The no-EPC index leaves the intended trajectory and then reverts, so nothing durable is priced.

Rent-increase cap

Textbook Grossman and Milgrom unravelling: undisclosed dwellings are penalised and sit at the energy-intensive level, while efficient property is €1.1/m² clear of both.

Tourist, vacancy tax

The sign flips. Non-disclosure earns a premium of €0.4 to €0.8/m², in line with Patras (2024, 2026).

No rent control

No significant price gradient at all. The EPC is not priced in, plausibly because too few renters here are EPC-sensitive.

One instrument, four regimes, three different signs. What disclosure does to price depends on the regime, and not in an order that tension alone explains: the most tense market of the four is the one where nothing durable is priced.

Model 2: First-Week Enquiries, Negative Binomial

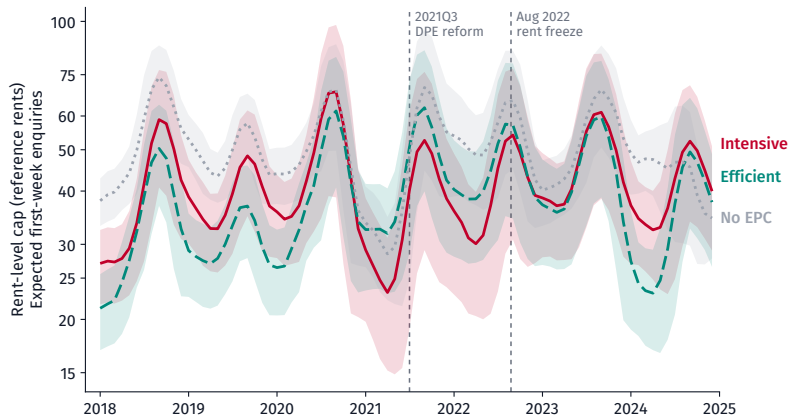
Model 2: first-week enquiries as an overdispersed count, with a live-day exposure offset

$$y_i \sim \text{NB2}(\mu_i, \phi), \quad \mu_i = \exp\left(\alpha + X_i' \beta + \sum_q \delta_q \mathbf{1}[q(t_i) = q] + \theta_{c(i)} + \underbrace{\log e_i}_{\text{offset}}\right)$$

Notation

- y_i : enquiries in the listing's first week
- $e_i = \text{clip}(\min(\text{tom}_i, 7), 1, 7)$: days the listing was actually live in that week, so the index reads **per live day** and short-lived listings are not mechanically penalised
- ϕ : dispersion, $\text{Var} = \mu + \phi\mu^2$, estimated per cell, $\hat{\phi} \in [0.73, 1.03]$

No significant demand gap between EPC labels, a feature of this highly tense market (rent-level cap)

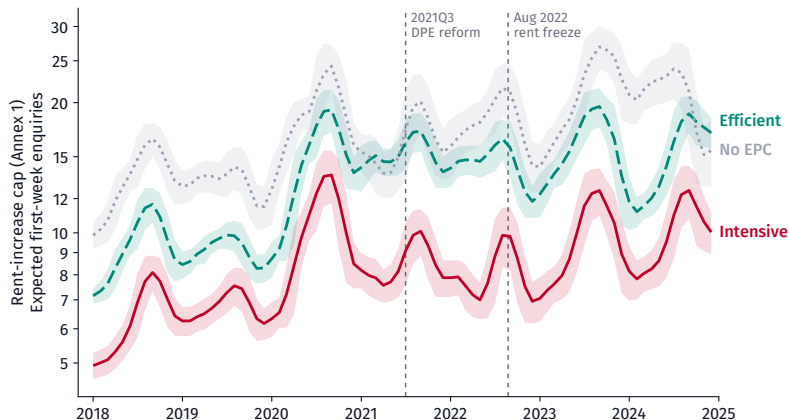


No-EPC level and gaps to it, first-week enquiries

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	48.7	51.8	49.4
Intensive	—9.1	—8.8	—5.3
Efficient	—13.8	—8.8	—8.8

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

Strong demand penalty on energy-intensive dwellings, and indifference between undisclosed and efficient (rent-increase cap)

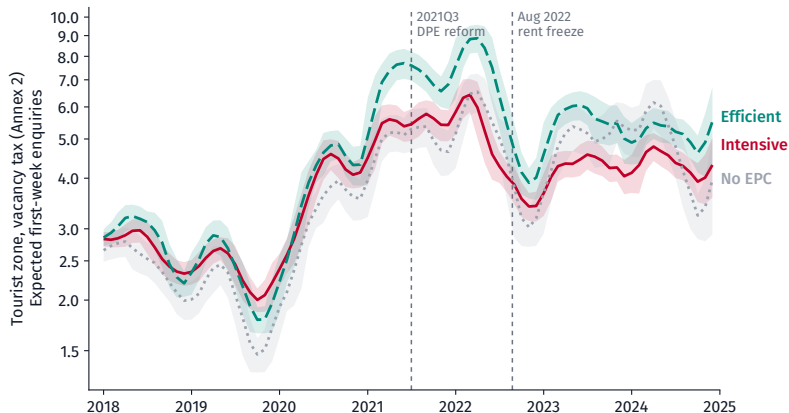


No-EPC level and gaps to it, first-week enquiries

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	14.9	19.6	20.3
Intensive	-7.2	-10.4	-10.8
Efficient	-3.5	-4.3	-5.1

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

No significant demand gap between EPC labels, a feature of this highly tense market (tourist zones)

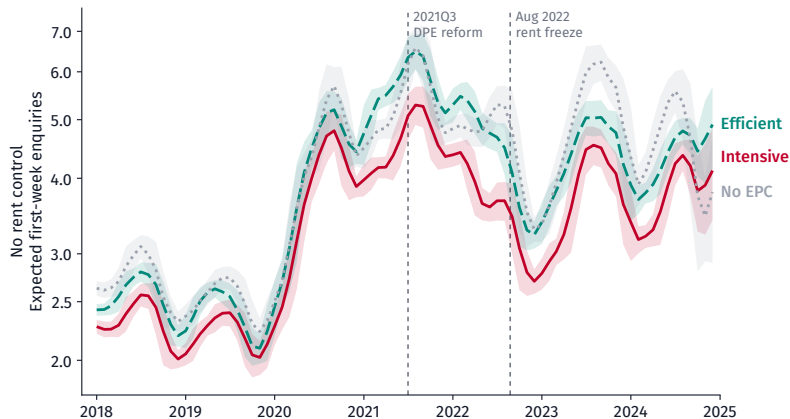


No-EPC level and gaps to it, first-week enquiries

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	2.9	4.9	4.6
Intensive	+0.4	-0.3	-0.4
Efficient	+0.7	+1.0	+0.5

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

Slight penalty on energy-intensive classifieds, and indifference between undiscovered and efficient (no rent control)

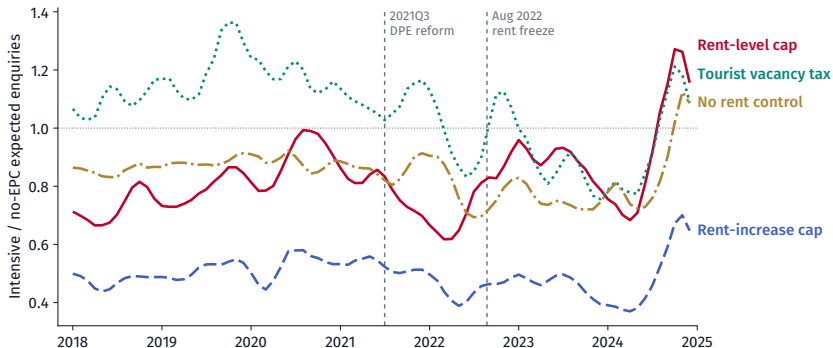


No-EPC level and gaps to it, first-week enquiries

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	3.4	4.9	4.6
Intensive	-0.4	-1.0	-1.0
Efficient	0.0	-0.2	-0.4

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

By 2024 only the rent-increase-cap zone still has no-EPC listings out-drawing intensive ones



Ratio of expected first-week enquiries, intensive over no-EPC. Below 1, a no-EPC listing out-draws an otherwise identical intensive one. Three regimes cross over the line during the window: the rent-level-cap zone runs from 0.70 in 2018Q1 to 1.23 by 2024Q4, the not-tight zone from 0.86 to 1.08. Only the rent-increase-cap zone stays below 1 throughout, at 0.67 in 2024Q4.

Takeaway: the demand penalty tracks the intensive label, and it is not ordered by tension

Where the label sorts seekers

Rent-increase cap: a strong penalty on energy-intensive dwellings, about 11 enquiries below no-EPC on a base of 20.

No rent control: the same sign, far smaller, about 1 enquiry below no-EPC on a base of 5.

In both, seekers are indifferent between undisclosed and efficient listings.

Where it does not

The rent-level-cap and tourist markets show no significant gap between any pair of labels.

Tension does not order this

The rent-increase-cap zone is a tense zone by definition (Annex 1), yet it behaves like the untight market while the other two tense regimes show nothing. Whatever separates the two groups, it is not how tight the market is. Two candidates worth testing: rents pinned to reference levels under the level cap, and thin, seasonal demand in tourist zones.

Where the penalty binds it falls on the disclosed bad rating, never on the silence: withholding the EPC costs nothing on the demand margin in any of the four markets.

Model 3: Time on Market, Discrete-Time Hazard

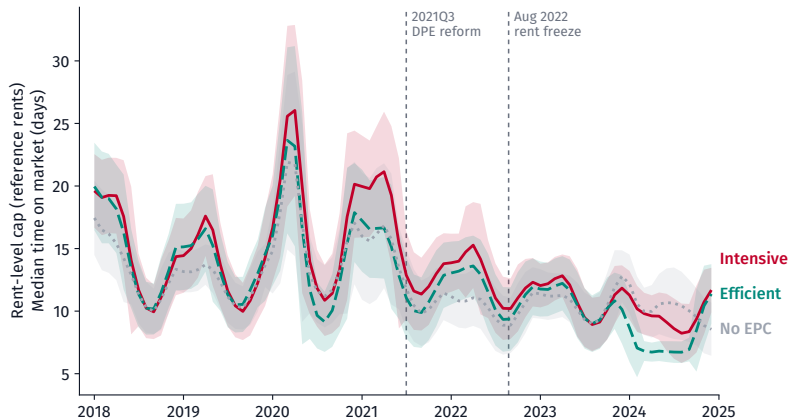
Model 3: time on market as a discrete-time hazard, one row per spell-week

$$\underbrace{\log(-\log(1 - h_{ik}))}_{\text{cloglog}} = \gamma_{a(k)} + \sum_q \delta_q \mathbf{1}[q(\text{mid}_{ik}) = q] + X_i' \beta + \theta_{c(i)}$$

Notation

- $h_{ik} = P(\text{exit in spell-week } k \mid \text{alive})$
- $\gamma_{a(k)}$: piecewise-constant baseline in spell age (0–1, 1–2, 2–4, 4–8, 8–26, 26+ **weeks**), so duration dependence is unrestricted
- mid_{ik} : midpoint of risk-week k , so the calendar axis reads “**the market** in quarter q ”, not the launch quarter
- Risk weeks run to $\lfloor p99/7 \rfloor = 56$

No significant difference in clearing pace between EPC labels (rent-level cap)

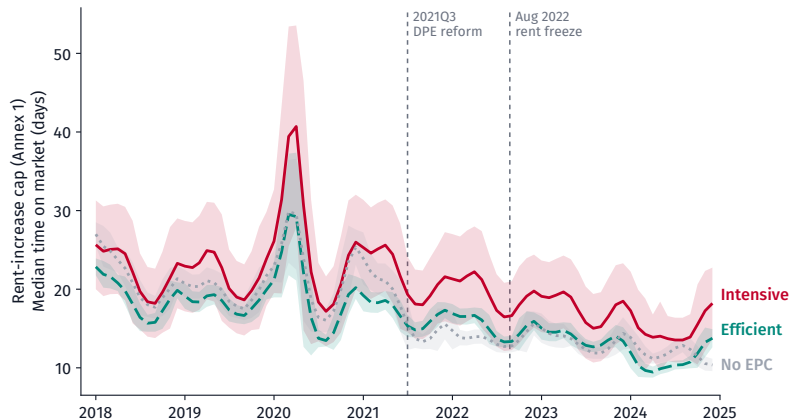


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	14.0	10.5	10.4
Intensive	+1.8	+0.9	+0.1
Efficient	+0.7	-0.3	-0.9

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

No significant difference in clearing pace between EPC labels (rent-increase cap)

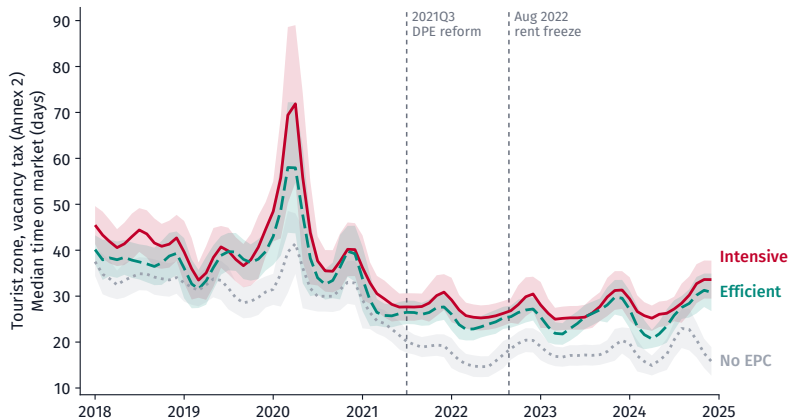


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	21.2	13.2	12.8
Intensive	+2.4	+4.5	+3.8
Efficient	-2.2	+0.5	-0.1

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

No significant difference in clearing pace between EPC labels (tourist zones)

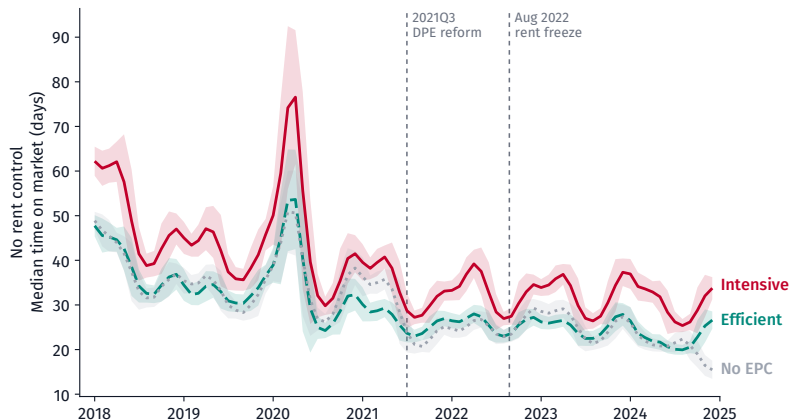


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	31.8	18.0	18.3
Intensive	+9.3	+9.9	+9.8
Efficient	+5.6	+7.6	+7.5

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

Energy-intensive classifieds clear significantly slower than the rest, consistent with Patras (2024, 2026)

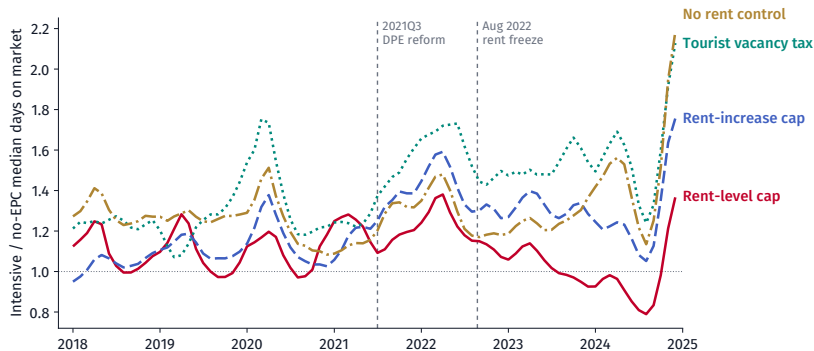


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	36.0	23.9	23.8
Intensive	+9.4	+7.8	+7.8
Efficient	-1.1	+0.8	+0.4

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

By 2024 intensive listings take 19 to 88% longer to let than no-EPC ones, in every regime



Ratio of median days on market, intensive over no-EPC. Window means run 1.10 to 1.42 over the plotted range, and every regime ends above 1.

Takeaway: clearing pace separates the bands only outside the tight zones

Three of the four markets

No significant difference in clearing pace between EPC labels in the rent-level-cap, rent-increase-cap and tourist markets, all three of them tight zones, whatever the point estimates suggest.

No rent control

Energy-intensive classifieds clear significantly slower than the rest, consistent with Patras (2024, 2026). This is the only market where the duration margin responds at all.

Only the market with no tight-zone designation separates the bands on clearing pace. All three tight-zone regimes show no difference, tourist zones included, even though they are the slowest of the four to let.

Model 4: Time on Market, Weibull AFT

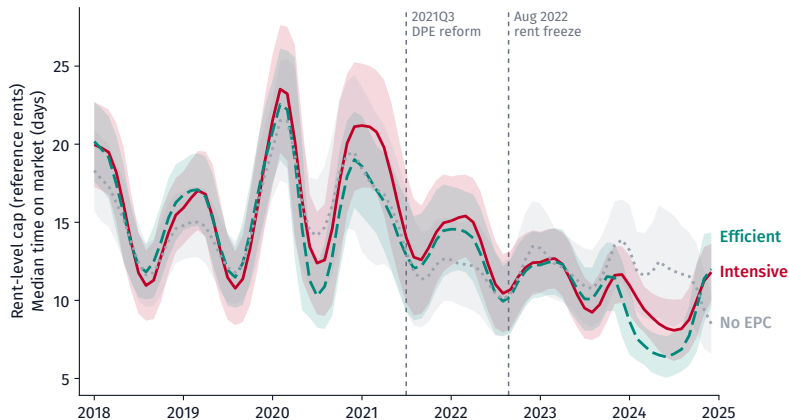
Model 4: the same duration made parametric, a Weibull AFT cross-check on Model 3

$$\log T_i = \underbrace{\alpha + \delta_{q(m_{0i})} + X_i' \beta + \theta_{c(i)}}_{\mu_i} + \sigma w_i, \quad w_i \sim \text{Gumbel}_{\min}$$

Notation

- T_i : duration in days, **one row per spell** (not per spell-week), same three-part censoring rule as Model 3
- m_{0i} : the listing's **start** month
- $p = 1/\sigma$: shape, with $p < 1$ a declining hazard and $p > 1$ rising. Fitted **0.69–0.90 in all 12 cells**, every one more than 2 SE below 1, so listings go stale rather than accelerating
- Median has a closed form, $e^{\mu}(\ln 2)^{\sigma}$

The AFT agrees on the rent-level-cap zone: the no-EPC speed advantage turns into a small deficit

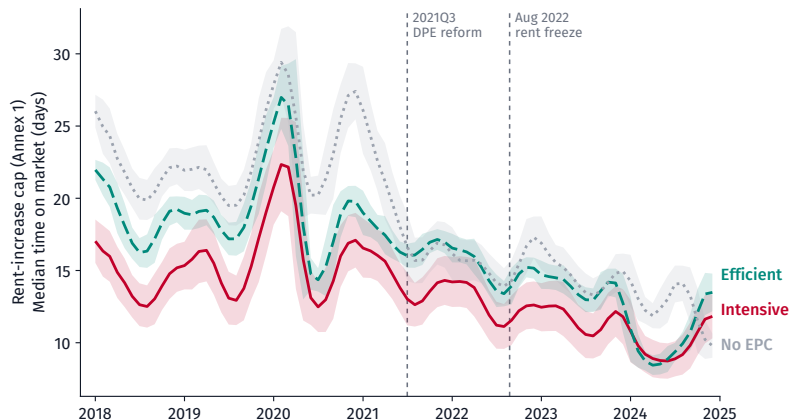


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	15.7	11.8	11.8
Intensive	+0.8	-0.3	-1.3
Efficient	+0.1	-0.9	-1.9

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

The one cell where the two duration models disagree: the AFT has intensive listings letting faster than no-EPC

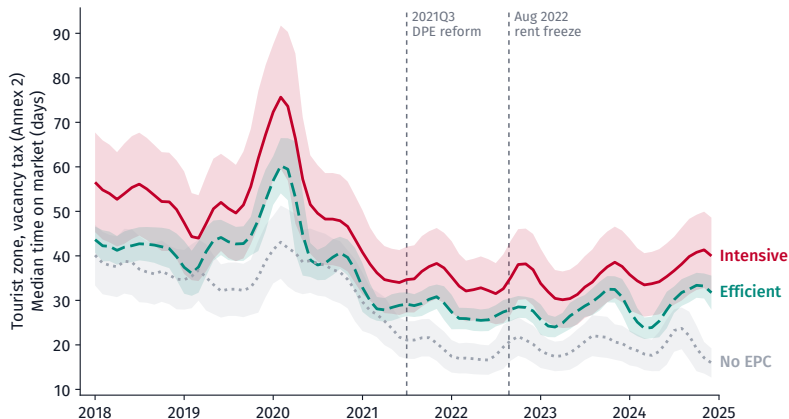


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	22.8	14.5	13.9
Intensive	-7.2	-2.8	-3.0
Efficient	-3.8	-0.9	-1.5

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

In tourist zones the AFT finds a much wider intensive gap than the hazard model: 16 days against 9

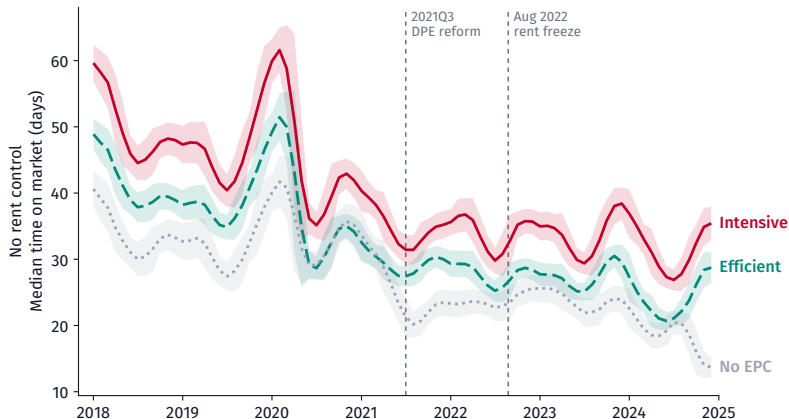


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	35.1	19.5	19.8
Intensive	+16.5	+15.7	+15.8
Efficient	+6.3	+8.7	+8.7

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

Outside tight zones the AFT puts both disclosed bands further behind no-EPC than the hazard model does

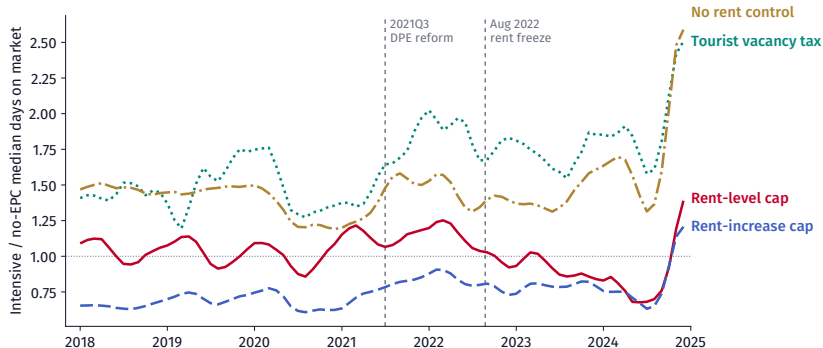


No-EPC level and gaps to it, median days on market

	Pre Jul '21	Post Jul '21	Post Aug '22
No EPC	32.9	22.0	21.7
Intensive	+13.0	+11.2	+11.3
Efficient	+5.1	+4.9	+4.5

Top row: the no-EPC band's own mean level. Below: each band's mean monthly gap to it. 2018Q1 to 2024Q4, figure units. The two post windows overlap, both running to the end.

The AFT tells the same non-disclosure story as the hazard, except in the rent-increase-cap zone



Ratio of median days, intensive over no-EPC, as window means. Tourist 1.64 and not-tight 1.46 against the hazard model's 1.42 and 1.30. But rent-increase-cap comes out at 0.75 against the hazard's 1.22, a sign flip, and it is the cell where the AFT's own median sits furthest from the raw one (ratio 0.53). Prefer Model 3 there.

Takeaway: the parametric cross-check leaves the market-level reading intact

Where the two models agree

Same sign and comparable magnitude in three of the four markets, with the AFT putting the disclosed bands somewhat further behind no-EPC in the tourist and not-tight markets.

Duration dependence

The shape parameter is below 1 in all 12 cells, every one by more than 2 SE: listings go stale, they never accelerate toward exit.

Where they disagree

Only the rent-increase-cap market, where the AFT has intensive listings letting faster than no-EPC and the hazard has them slower. That is also the cell whose modelled median sits furthest from the raw one.

Which to read

Model 3 stays headline: it is nonparametric in duration and reads the market in each quarter, where the AFT reads the quarter a listing was launched.

No market-level conclusion turns on the choice between the two duration models, except in the rent-increase-cap zone, where the hazard is the safer read.

Thank You

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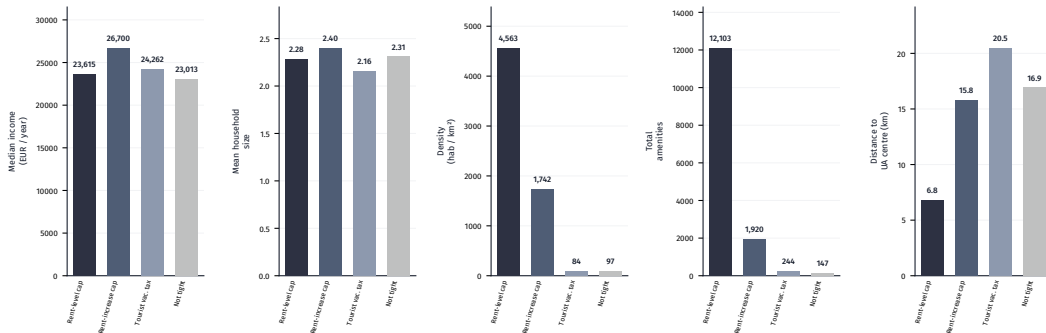
SciencesPo.

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- . 2018. *Loi no. 2018-1021 du 23 novembre 2018 portant évolution du logement, de l'aménagement et du numérique (ELAN)*. Journal officiel de la République française no. 0272, art. 140. Rent-level control (*encadrement des loyers*) in the tightest zones, November.

Unweighted commune means: the rent-increase-cap zone is richest when Paris and Lyon do not dominate

Commune composition by tension zone (unweighted means)



Unweighted commune means (each commune counts equally). Source: FiloSoFi 2021, BPE 2025, AAV 2020, IGN.